

Q – What's the difference between Linux and UNIX?

Q – What is a kernel?

UNIX Operating System

- It was publicly introduced on 1st Jan 1970.
- Developed by Dennis Ritchie and Ken Thompson in AT & T Bell Labs.
- Its developed using "C" language.
- It basically provides the concept to work in any Linux/UNIX platform.
- UNIX = Uniplexed Information and Computing System (UNICS → UNIX)
- freeware

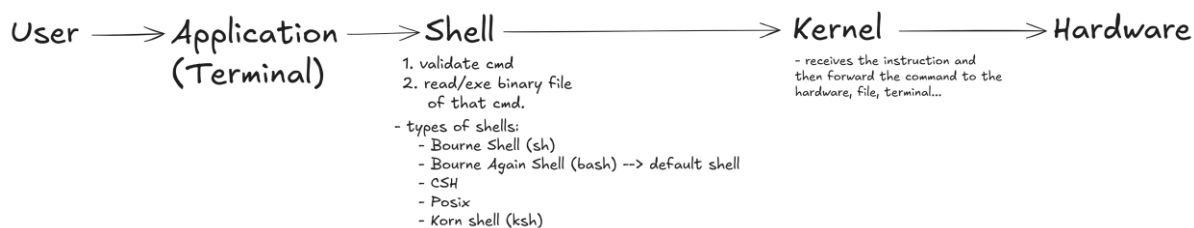
Linux Operating System

- It was introduced on 17th Sept 1991.
- Developed by Linus Benedict Torvalds.
- It is derived from UNIX operating system.
- Unlike UNIX, Linux is Hardware independent.
- Opensource

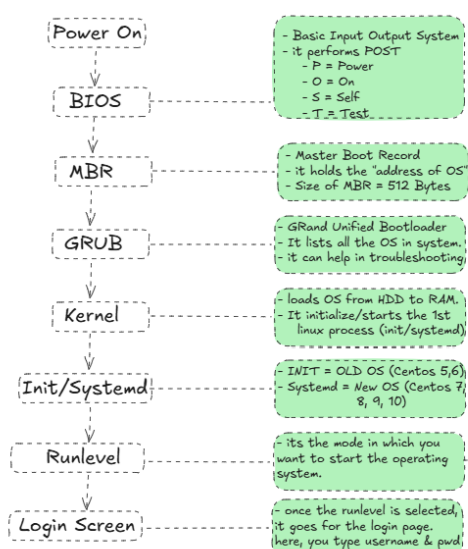
Opensource – free to download, free to use and free to share, you can edit the "source code".

Freeware – free to download, free to use and free to share, you cannot edit the source code of an OS.

Linux Architecture



Booting process:



Runlevel	MEANING	NEW CMD
init 0	halt/shutdown	poweroff.target
init 1	Single user maint. mode	rescue.target
init 2	Multi user maint. mode	multi-user.target
init 3	full CLI mode	multi-user.target
init 4	unused	multi-user.target
init 5	init 3 + GUI	graphical.target
init 6	reboot	reboot.target

→ Current Runlevel

File system Hierarchy standard:

- This helps in understanding the file system architecture.
- It starts from "/" → forward slash.
- "/" contains approx. 14 directories (folders)
 - */bin* – essential command binaries for REGULAR user.
 - */sbin* – command binaries for ROOT user.
 - */boot* – bootable files in it. Minimum size required: 500MB (350MB recommended).
 - */dev* – device (hardware) related files.
 - */etc* – contains all the configuration related file {IMP}.
 - */home* – user's home directory.
 - */lib* – library for developers
 - */media* – for media files like CD/DVD
 - */mnt* – mount point for temporary filesystems (pen drive).
 - */opt* – additional software packages installed by user.
 - */proc* – interface to kernel data structures, running processes also.
 - */root* – Root user's home directory
 - */tmp* – for temporary files.
 - */usr* – UNIX system resources
 - */var* – log files (boot, yum, rpm, security, login...)